**جامعة بورسعيد**
**كلية الصيدلة**
**قسم الأدوية والسموم**

# بروتوكول

## رسالة الماجيستير الخاصة بالصيدلي/ رفعت إبراهيم رفعت

المعيد بقسم الأدوية والسموم - كلية الصيدلة - جامعة بورسعيد - مصر

## بعنوان

استهداف التهاب المفاصل المستحث تجريبيًا في الجرذان باستخدام بعض النواتج الطبيعية كعامل مقلل للاعتماد على الجلوكوكورتيكويدات.

**خطة بحثية مقدمة لكلية الصيدلة تمهيدًا للحصول على درجة الماجيستير في العلوم الصيدلية**

**(الأدوية والسموم)**

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تحت إشراف

**أ.م.د/ أحمد أبو الفتوح السيد مراد**

رئيس القسم، أستاذ مساعد الأدوية والسموم، كلية الصيدلة - جامعة بورسعيد

**أ.م.د/ أحمد جعفر أحمد**

أستاذ مساعد الأدوية والسموم، كلية الصيدلة - جامعة بورسعيد

**أ.م.د/ أحمد سعد سعد**

أستاذ مساعد الأدوية والسموم، كلية الصيدلة - جامعة بورسعيد

**كلية الصيدلة**

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**يعتمد**

**عميد الكلية**

**Port Said University**
**Faculty of Pharmacy**
**Department of pharmacology and toxicology**

## A protocol for a master's degree in

## Department of pharmacology and toxicology

Presented by

**Riffat Ibrahim Riffat**

## Thesis title

Targeting Experimentally Induced Arthritis in Rats Using Selected Natural Products as Glucocorticoid-Sparing Agents.

Under supervision of

**Associate Professor/ Ahmed Aboul-Fotouh Elsayed Mourad**

Head of the department, Assistant Professor of Pharmacology and Toxicology, Faculty of Pharmacy, Port-Said University

**Associate Professor/ Ahmed Gaafar Ahmed**

Assistant Professor of Pharmacology and Toxicology, Faculty of Pharmacy, Port-Said University

**Associate Professor/ Ahmed Saad Saad**

Assistant Professor of Pharmacology and Toxicology, Faculty of Pharmacy, Port-Said University

**Port Said University**

**2026**

**Dean of the Faculty**

## Introduction

Rheumatoid arthritis (RA) is a chronic autoimmune disease characterized by persistent synovial inflammation and joint destruction, driven by a complex network of pro-inflammatory cytokines and intracellular signaling cascades like NF-kB and MAPK. Glucocorticoids are highly effective for RA management but are associated with severe long-term adverse effects, creating a need for glucocorticoid-sparing agents. Recent studies into various natural products have identified compounds which may potentiate the anti-inflammatory effects of low-dose glucocorticoids via PPAR-alpha and glucocorticoid receptor-alpha crosstalk.

## Aim of the work

This study aims to investigate the potential of selected natural products as a glucocorticoid-sparing agent, and to evaluate the efficacy of this combination against established antirheumatic agents in adjuvant-induced arthritis.

## Study Design

This study employs the Complete Freund's Adjuvant (CFA)-induced arthritis model in female Wistar rats to investigate the glucocorticoid-sparing potential of micronized palmitoylethanolamide (m-PEA), both alone and in combination with prednisolone, relative to the reference agents baricitinib and tacrolimus. Arthritis is induced on Day 0 by a single subcutaneous injection of 0.3 mL CFA into the right inguinal/proximal thigh region, while negative control animals receive an equivalent subcutaneous injection of 0.25 mL paraffin oil. Following a 15-day induction period, oral treatment is initiated on Day 16 and administered once daily until Day 45, after which all animals are sacrificed on Day 46 for terminal sample collection.

Animals are allocated to nine experimental groups [number of rats per group not specified]: a negative control group, a positive (CFA arthritic) control group, a prednisolone monotherapy group (10 mg/kg), an m-PEA monotherapy group (200 mg/kg), three PEA/prednisolone combination groups at ascending PEA doses (50/5, 100/5, and 200/5 mg/kg), a tacrolimus reference group (1.8 mg/kg), and a baricitinib reference group (10 mg/kg). This design permits evaluation of whether escalating doses of m-PEA in combination with a fixed low dose of prednisolone can achieve comparable therapeutic efficacy to full-dose prednisolone monotherapy or to the established reference agents, thereby addressing the study's glucocorticoid-sparing objective.

## Experimental Design Table

| Group | Induction/Control Condition | Treatment | Dose | Route | Treatment Period | Purpose |
|---|---|---|---|---|---|---|
| Negative control | 0.25 mL paraffin oil, subcutaneous, Day 0 | None | — | — | Day 16–Day 45 | Normal physiological reference |
| Positive control (CFA arthritic) | 0.3 mL CFA, subcutaneous, Day 0 | None (vehicle) | — | Oral | Day 16–Day 45 | Untreated arthritis reference |
| Prednisolone alone | CFA-induced | Prednisolone | 10 mg/kg | Oral | Day 16–Day 45 | Full-dose glucocorticoid comparator |
| m-PEA alone | CFA-induced | m-PEA | 200 mg/kg | Oral | Day 16–Day 45 | Evaluation of m-PEA monotherapy efficacy |
| PEA/Prednisolone 50/5 | CFA-induced | m-PEA + Prednisolone | 50/5 mg/kg | Oral | Day 16–Day 45 | Low-dose PEA glucocorticoid-sparing combination |
| PEA/Prednisolone 100/5 | CFA-induced | m-PEA + Prednisolone | 100/5 mg/kg | Oral | Day 16–Day 45 | Intermediate-dose PEA glucocorticoid-sparing combination |
| PEA/Prednisolone 200/5 | CFA-induced | m-PEA + Prednisolone | 200/5 mg/kg | Oral | Day 16–Day 45 | High-dose PEA glucocorticoid-sparing combination |
| Tacrolimus | CFA-induced | Tacrolimus | 1.8 mg/kg | Oral | Day 16–Day 45 | Calcineurin inhibitor reference comparator |
| Baricitinib | CFA-induced | Baricitinib | 10 mg/kg | Oral | Day 16–Day 45 | JAK inhibitor reference comparator |

## Measured Parameters

Disease activity and underlying molecular mechanisms are assessed using serum and knee joint tissue samples collected at sacrifice. Serum samples are analyzed by ELISA for anti-citrullinated protein antibodies (ACPA) and the pro-inflammatory cytokines TNF-α, IL-1β, and IL-6. Knee joints are processed for histopathological examination to assess structural joint changes, and for immunohistochemical evaluation of NF-κB p65 nuclear expression and TLR4 expression as markers of inflammatory signaling activation. Additionally, knee joint tissue homogenates are analyzed by ELISA for p38 MAPK expression to further characterize the intracellular inflammatory signaling pathways involved.

## Measured Parameters Table

| Category | Parameter | Sample/Tissue | Method | Relevance |
|---|---|---|---|---|
| Serum ELISA | ACPA | Serum | ELISA | Autoantibody marker associated with arthritis |
| Serum ELISA | TNF-α | Serum | ELISA | Systemic pro-inflammatory cytokine |
| Serum ELISA | IL-1β | Serum | ELISA | Systemic pro-inflammatory cytokine |
| Serum ELISA | IL-6 | Serum | ELISA | Systemic pro-inflammatory cytokine |
| Histopathology | Joint structural changes | Knee joint | Histopathological examination | Structural assessment of joint inflammation and damage |
| Immunohistochemistry | NF-κB p65 nuclear expression | Knee joint | Immunohistochemistry | Marker of inflammatory transcriptional signaling activation |
| Immunohistochemistry | TLR4 expression | Knee joint | Immunohistochemistry | Marker of innate immune inflammatory signaling |
| Tissue ELISA | p38 MAPK | Knee joint homogenate | ELISA | Intracellular inflammatory signaling pathway marker |

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